

# Reproduction: Stealing ML Models via Prediction APIs

Python 3 local black-box reproduction

Model-Stealing-Reproduce-Pre

# Reproduction Setting

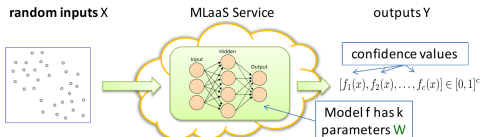
## Local black-box oracle

- Train a target model  $f$  locally.
- Expose confidence, label, or leaf-id queries.
- Recover a clone  $f'$  without reading weights.

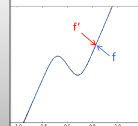
## Reproduction choice

Python 3 local oracle; same attack interface, no AWS/BigML account or legacy Python 2 stack.

## Generic Equation-Solving Attacks



- Solve **non-linear equation system** in the weights  $W$ 
  - Optimization problem + gradient descent
  - *"Noiseless Machine Learning"*
- Multinomial Regressions & Deep Neural Networks:
  - **>99.9% agreement between  $f$  and  $f'$**
  - $\approx 1$  query per model parameter of  $f$
  - 100s - 1,000s of queries / seconds to minutes



## Experiments and Interfaces

| Attack                      | Dataset       | Target model              | Black-box response |
|-----------------------------|---------------|---------------------------|--------------------|
| Binary equation-solving     | Breast Cancer | Logistic Regression       | confidence         |
| Multiclass equation-solving | Iris / Digits | Softmax / OvR LR          | confidence         |
| Confidence rounding         | Digits        | Softmax / OvR LR          | rounded confidence |
| Decision-tree path finding  | Iris          | DecisionTree(max_depth=3) | leaf id            |
| Label-only retraining       | Digits        | Softmax LR                | class label        |
| KLR leakage visualization   | Digits        | RBF representer model     | extracted params   |

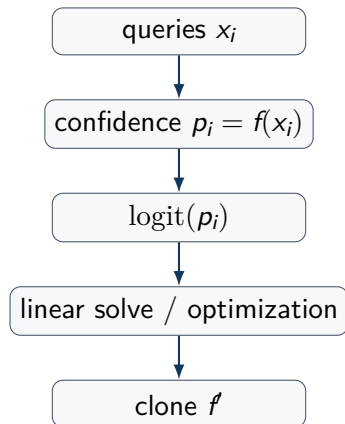
### Metrics

- **Test agreement:** fraction of test inputs where  $f'(x) = f(x)$ .
- **Uniform agreement:** agreement on random points sampled from the normalized feature box.
- **TV distance:** mean total variation distance between returned probability vectors.

## Binary Logistic Regression

$$p = \sigma(w^\top x + b), \quad \log \frac{p}{1-p} = w^\top x + b$$

- Breast Cancer has  $d = 30$  features.
- Unknowns are  $w_1, \dots, w_{30}, b$ .
- $d + 1 = 31$  independent confidence queries give a linear system.
- Solving  $[X, 1]\theta = \text{logit}(p)$  recovers the model.

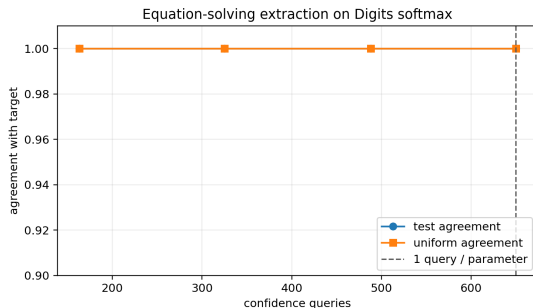


# Equation-Solving Results

| Model          | Params | Queries | Test | Uniform |
|----------------|--------|---------|------|---------|
| Binary LR      | 31     | 31      | 100% | 100%    |
| Iris softmax   | 15     | 15      | 100% | 100%    |
| Digits softmax | 650    | 650     | 100% | 100%    |
| Digits OvR     | 650    | 650     | 100% | 100%    |

## Observation

650 confidence queries extract the Digits softmax model, matching the paper's one-query-per-parameter behavior.

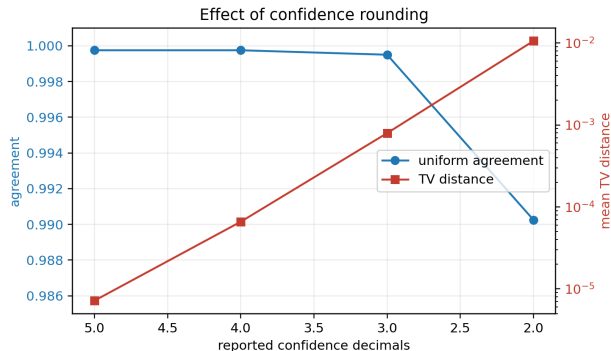


# Confidence Rounding

| Decimals | Queries | Test    | Uniform |
|----------|---------|---------|---------|
| 5        | 650     | 100.00% | 99.97%  |
| 4        | 650     | 100.00% | 99.96%  |
| 3        | 650     | 100.00% | 99.90%  |
| 2        | 650     | 99.63%  | 99.09%  |

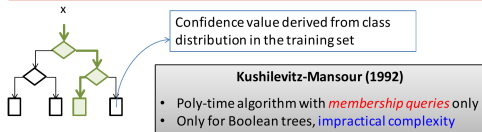
## Result

Four or five decimal places remain enough for extraction. Rounding to two or three decimals increases probability error but does not remove the attack surface.



# Decision Trees: Differential Testing with Leaf IDs

## Extracting a Decision Tree



### (Ab)using Confidence Values

- Assumption: all tree leaves have *unique confidence values*
- *Reconstruct tree decisions* with “differential testing”
- Online attacks on BigML



Inputs  $x$  and  $x'$  differ in a single feature



## Path-finding in the reproduction

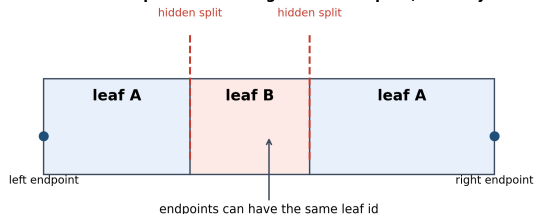
- 1 Query a point and record its leaf id.
- 2 Change one feature at a time.
- 3 A changed leaf id indicates a split on that feature.
- 4 Use binary search for continuous thresholds.
- 5 Recurse on newly discovered regions.

## Result on Iris

Target leaves: 5; extracted leaves: 5; queries: 557; test and uniform agreement: 100%.

# Decision Tree Implementation Detail

**Implementation note: recover predicates along the current path, not only interval endpoints**



## Why the naive scan fails

If only the two interval endpoints are checked, both may return the same leaf id while an unseen middle region reaches a different leaf.

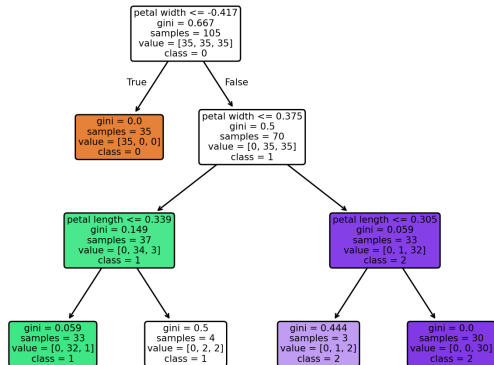
## Implementation choice

The extractor recovers predicates along the current path and tracks interval constraints per feature. This avoids missing hidden leaves and handles  $\leq$  /  $>$  threshold boundaries explicitly.



# Extracted Tree: Larger View

Target sklearn tree



Extracted leaf intervals

```
leaf 1 -> class 0
  x3: [-1.00, -0.42]

leaf 4 -> class 1
  x2: [-1.00, 0.34]; x3: [-0.42, 0.38]

leaf 5 -> class 1
  x2: [0.34, 1.00]; x3: [-0.42, 0.38]

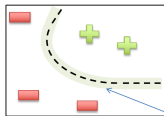
leaf 7 -> class 2
  x3: [0.38, 1.00]

leaf 8 -> class 2
  all features
```

# Label-Only Extraction

## Generic Model Retraining Attacks

- Extend the Lowd-Meek approach to non-linear models
- **Active Learning:**
  - Query points close to “decision boundary”
  - Update  $f'$  to fit these points
- Multinomial Regressions, Neural Networks, SVMs:
  - >99% agreement between  $f$  and  $f'$
  - $\approx 100$  queries per model parameter off



$\approx 100\times$  less efficient  
than equation-solving

query more  
points here

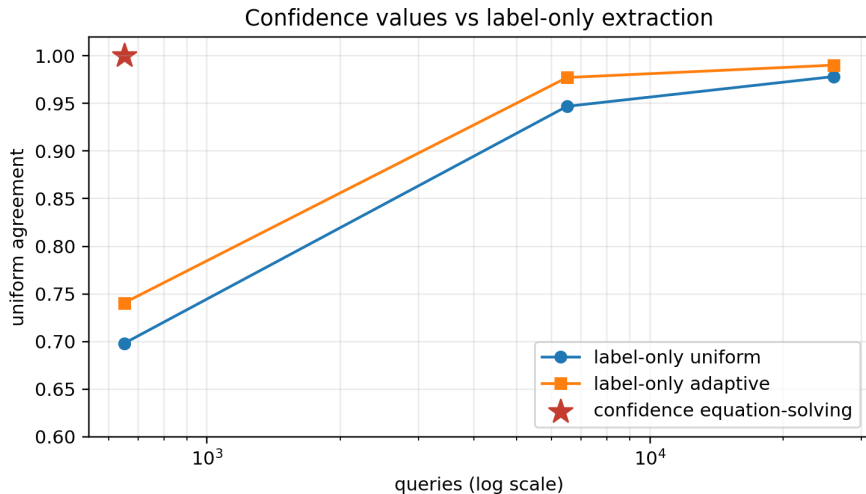
Source: Tramèr et al., USENIX Security 2016 slides, p.16

| Method           | Queries | Uniform agreement |
|------------------|---------|-------------------|
| confidence solve | 650     | 100.00%           |
| label uniform    | 650     | 68.83%            |
| label adaptive   | 650     | 76.13%            |
| label adaptive   | 6,500   | 97.83%            |
| label adaptive   | 26,000  | 99.33%            |

## Cost shift

Returning labels only removes the equation-solving shortcut. Adaptive retraining still approaches the target, but needs roughly two orders of magnitude more queries.

# Confidence vs Label-Only Query Cost



# KLR Leakage: Parameters Can Contain Training Samples

## Representer model used for visualization

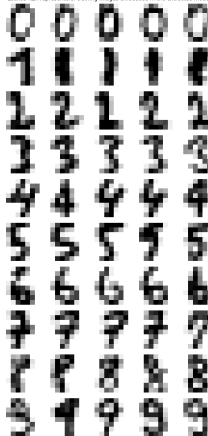
$$s_c(x) = \sum_{r: y_r=c} \exp(-\gamma \|x - r\|^2)$$

Digits dataset; 5 representers per class; 50 extracted images in total.

## Security implication

Extraction may reveal more than functionality. For representer-style kernel models, recovered parameters can directly expose training examples.

Leaked KLR representers: training images embedded in the extracted model



# Reproduction Takeaways

## What was reproduced locally

- Confidence APIs make LR extraction a system-solving problem.
- Multiclass LR reaches perfect agreement at roughly one query per parameter.
- Decision-tree extraction depends on leaf identity and careful path predicates.
- Label-only APIs raise query cost but do not eliminate extraction.
- Representer-style kernel models illustrate data leakage risk.

## Boundary of this reproduction

AWS/BigML online experiments were not rerun; the goal is a local, auditable reproduction of the attack mechanisms for presentation.